



BASIC ELECTRICAL AND ELECTRONICS ENGINEERING LAB & SIMULATION

DEPARTMENT OF ELECTRICAL ENGINEERING

EXPERIMENT NO.: 1

EXPERIMENT NAME: Study and Identification of Electrical Components, Measuring Instruments and Safety Devices.

OBJECTIVE: To identify and understand the functions of various basic electrical components, measuring instruments and safety devices, and to learn their safe and correct use in electrical laboratory applications.

APPARATUS REQUIRED:

Sl. No.	Apparatus/Components	Quantity
1.	Resistors (Carbon/Wire wound)	Various
2.	Capacitors (Electrolytic/Ceramic)	Various
3.	Inductors	Various
4.	Switches (SPST, SPDT)	Various
5.	ICDP	1
6.	Fuse	1
7.	MCB	1
8.	RCCB/ELCB	1 each
9.	Digital Multimeter (DMM)	1
10.	Clamp Meter	1
11.	Voltmeter(Analog & Digital)	1 each
12.	Ammeter(Analog & Digital)	1 each
13.	Wattmeter(Analog & Digital)	1 each
14.	Megger (Insulation Tester)	1
15.	Energy Meter	1

THEORY:

Electrical laboratories contain various passive and active components, measuring instruments and protective devices. Resistors limit current, capacitors store electric charge, inductors store magnetic energy, diodes allow unidirectional current flow, while switches control circuits.

Measuring instruments such as voltmeters, ammeters, multimeters, clamp meters, meggers and earth testers are used for measuring electrical quantities and testing equipments.

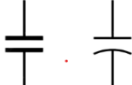
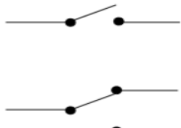
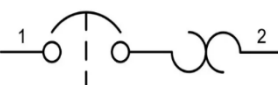
Safety devices including fuses, MCBs, RCCBs, relays and proper earthing protect personnel and equipment from overloads, short circuits and leakage currents.

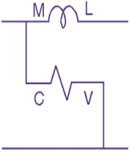
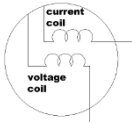
PROCEDURE:

1. Observe each electrical component.
2. Identify its symbol, rating and application.
3. Study each measuring instrument and its terminals.
4. Observe safety devices and understand their operation.
5. Record observations in the observation table.

OBSERVATION:

Prepare tables for electrical components, measuring instruments, and safety devices with columns for Name, Rating/Range, Symbol and Applications.

Sl. No.	Components/ Instruments	Symbol	Rating/Range	Applications
1.	Resistors (Carbon/ Wire wound)		Carbon: 10 Ω –1 M Ω , 0.25–0.5 W Wire-wound: 1 Ω –10 k Ω , 5–20 W	1) Current limiting; 2) Voltage division.
2.	Capacitors (Ceramic/ Electrolytic)		Ceramic: 100 pF–1 μ F, 50 V Electrolytic: 1–1000 μ F, 25/50 V	1) Filtering/smoothing; 2) Coupling or timing.
3.	Inductors		1 mH–1 H; 0.1–1 A (typ.)	1) Choke/filter; 2) Energy storage.
4.	Switches (Single Pole Single Throw- SPDT, Single Pole Double Throw- SPDT)		SPST/SPDT; 6 A, 250 V AC	1) Circuit ON/OFF; 2) Circuit selection.
5.	Iron Clad Double Pole (ICDP)		DP, 32 A, 240 V AC, 50 Hz	1) Main isolation; 2) Load switching.
6.	Fuse		5/10 A, 250 V AC	1) Overcurrent protection; 2) Short-circuit protection.
7.	MCB		SP, C6/C10; 230 V AC; 6 kA	1) Overload protection; 2) Short-circuit protection.

8.	Residual Current Circuit Breaker (RCCB) / Earth Leakage Circuit Breaker (ELCB)		RCCB: DP, 25/40 A, 30 mA, 230 V AC ELCB: 230 V AC, 50 V trip (typ.)	1) Earth-leakage protection; 2) Electric-shock risk reduction.
9.	Digital Multimeter (DMM)	-	DC/AC V: 0 to 600 V; I: 0 to 10 A; R: to 20 M Ω	1) V-I-R measurement; 2) Continuity/component testing.
10	Clamp Meter	-	AC current: 0 to 600 A; AC/DC voltage: 0 to 600 V	1) Current without opening circuit; 2) Load troubleshooting.
11	Voltmeter (Analog & Digital)		Analog: 0–300 V AC; 0–30 V DC Digital: 0–600 V AC/DC	Analog: 1) Panel indication; 2) Trend observation. Digital: 1) Precise voltage; 2) Circuit testing.
12	Ammeter (Analog & Digital)		Analog: 0–5 A AC/DC Digital: 0–10 A AC/DC	Analog: 1) Panel indication; 2) Load-current study. Digital: 1) Precise current; 2) Circuit testing.
13.	Wattmeter (Analog & Digital)		Analog: 150/300 V, 5 A, to 1.5 kW Digital: to 600 V, 10 A, 2 kW	Analog: 1) Power indication; 2) Load experiment. Digital: 1) Precise power; 2) Efficiency testing.
14.	Megger (Insulation Tester)		Test: 250/500/1000 V DC; IR: to 200 G Ω	1) Cable insulation test; 2) Motor/winding insulation test.
15.	Energy Meter		1-phase, 230 V, 5–20 A, 50 Hz; 1600 imp/kWh	1) Energy consumption; 2) Load-energy experiment.

LABORATORY SAFETY RULES:

1. Never touch live conductors.
2. Wear insulated footwear.
3. Keep hands dry.
4. Switch OFF supply before making connections.
5. Use insulated tools.
6. Never overload circuits.
7. Ensure proper earthing.
8. Report damaged equipment immediately.

RESULT:

The various electrical components, measuring instruments, and safety devices were successfully identified and studied. Their functions, operating principles, ratings, and applications were understood, along with the essential safety precautions required for electrical laboratory work.

VIVA-VOCE QUESTIONS:

1. What is the function of a resistor?
2. Why is an ammeter connected in series?
3. Why is a voltmeter connected in parallel?
4. What is the purpose of an MCB?
5. What is RCCB?
6. Why is earthing necessary?
7. What is insulation resistance?
8. What is the function of a relay?
9. What is a clamp meter?
10. State five laboratory safety rules.